Beyond ‘Magic Numbers’: A Threshold Justification Stack for Clinical AI Governance Walter C. Brown, MSc, MA Ethical Alpha Audit, London, United Kingdom Correspondence: walter.brown@ethicalalphaaudit.com ┃ ORCID: 0000-0002-6050-8522 Article Type: Methodology and Theory ┃ Target Journal: BMJ Health & Care Informatics What is already known on this topic Clinical AI governance frameworks operationalise safety through quantitative thresholds, but the methodological rationale behind specific numeric values is rarely documented, producing “magic numbers” whose governance authority derives from convention rather than evidence. What this study adds This paper identifies six structural failure mechanisms in threshold design through conceptual synthesis and regulatory analysis, and proposes a tiered Threshold Justification Stack (TJS) as a normative governance proposal.

Primary Safety Thresholds require all six documentation layers.

Secondary Operational Thresholds require three. Full worked examples for both tiers are provided in the supplementary appendices.

The TJS is positioned as a structured augmentation of existing hazard log and clinical risk management processes under DCB0129/0160, not a replacement for them. How this study might affect research, practice or policy Auditors and governance committees gain a proposed proportionate documentation architecture for threshold justification, designed for integration into existing clinical safety case processes, audit software, EHR governance modules, and clinical AI oversight committee workflows. A proposed pilot design with prespecified feasibility endpoints—including median completion time per threshold record and inter-rater agreement (kappa) for tier classification—provides the pathway for empirical validation. Abstract Objectives: To characterise structural failure mechanisms in threshold design for clinical AI governance and to propose a tiered Threshold Justification Stack (TJS) as a governance proposal with full worked examples for both threshold tiers. Methods: Conceptual synthesis and regulatory analysis.

Candidate failure mechanisms were identified through targeted literature searches of PubMed, Web of Science, and regulatory document repositories (search dates: January 2018–December 2025), supplemented by citation tracking from key references. Inclusion criterion: the failure mechanism must be structural—arising from how thresholds are designed or deployed—rather than from errors in any individual numeric value. Candidate mechanisms were grouped by structural similarity and assessed for regulatory alignment against the EU AI Act (Articles 9, 10, 17, 72, Annex IV), FDA Predetermined Change Control Plan (PCCP) guidance, FUTURE-AI consensus guideline, WHO AI ethics guidance, and ISO/IEC 42001. Worked examples for both tiers are provided in the supplementary appendices. Results: Six structural failure mechanisms were identified: proxy thresholding, context collapse, threshold coupling, boundary gaming, epistemic asymmetry, and audit–lifecycle disconnect. A tiered TJS is proposed: Primary Safety Thresholds (failure causes direct patient harm) require all six layers; Secondary Operational Thresholds require three core layers.

The TJS is advanced as a structured proposal; its adoption at scale requires empirical feasibility assessment. Conclusions: The tiered TJS offers a proposed proportionate documentation architecture for threshold justification. The framework requires empirical validation through an inter-rater reliability exercise and committee feasibility pilot—with prespecified primary endpoint of median completion time per threshold record and secondary endpoint of inter-rater agreement kappa for tier classification—before claims of scalable applicability can be made.

Keywords: artificial intelligence; clinical governance; algorithmic auditing; threshold justification; medical informatics; patient safety; audit software integration INTRODUCTION

Clinical AI systems are increasingly deployed across the care continuum [1–3]. Governance frameworks operationalise safety and fairness through quantitative thresholds [4,5]. These values function as deployment decision criteria: cross the threshold and a system may be rejected, remediated, or flagged for re-audit. Numeric cutoffs frequently appear in audit checklists without documented justification, as though their appropriateness were self-evident—what commentators have described as “magic numbers” [14]: values whose governance authority rests on convention rather than evidence linking them to the harms they are intended to prevent.

The FUTURE-AI consensus guideline [26], while defining 30 best practices for trustworthy AI in healthcare, does not specify how the quantitative thresholds operationalising those practices should be justified or documented. Similarly, the FDA’s January 2025 draft guidance on AI-enabled device software functions [27] articulates total product lifecycle requirements but does not mandate a specific threshold documentation standard. When the basis for a threshold is opaque, audit credibility erodes regardless of how carefully other framework elements have been designed. A further challenge is proportionality. Requiring the same documentation depth for a calibration floor on a treatment-recommendation algorithm and for an alert-volume cap on an administrative tool imposes costs disproportionate to risk. This paper addresses both the quality gap (undocumented thresholds) and the proportionality gap (uniform requirements regardless of risk level) through a tiered framework.

The TJS is classified explicitly as a normative governance proposal with a tiered structure proportionate to threshold risk class. It synthesises existing best practice from safety engineering, regulatory science, and algorithmic auditing into a single documentation architecture. Its adoption at scale requires empirical feasibility assessment; this paper identifies the pilot design for that assessment. METHODS: CONCEPTUAL SYNTHESIS AND REGULATORY

ANALYSIS This study applies conceptual synthesis and regulatory analysis rather than systematic review. Search strategy and sources Targeted searches were conducted in PubMed, Web of Science, and Google Scholar using combinations of the terms “clinical AI threshold,” “algorithmic audit,” “governance metric,” “safety boundary,” “medical device threshold,” and “performance cutoff.” Regulatory document repositories searched included EUR-Lex (EU AI Act), FDA.gov (SaMD and PCCP guidance), MHRA.gov.uk, WHO institutional repository, and ISO online browsing platform. Search dates: January 2018–December 2025.

Citation tracking from key references (Leveson 2011, Obermeyer et al 2019, Thomas & Uminsky 2022, Lekadir et al 2025) supplemented the database searches. Inclusion criterion A candidate failure mechanism must be structural—arising from how thresholds are designed or deployed—rather than from errors in any individual numeric value. Grouping and synthesis process Candidate mechanisms identified from the literature were grouped by structural similarity. The grouping was performed by the author; two mechanisms were merged (initial candidates for “proxy bias” and “causal opacity” were consolidated into “proxy thresholding” on the grounds that both describe the same structural mechanism: a disconnect between the measured metric and the harm it purports to control). The resulting six mechanisms were assessed against existing governance instruments to identify documentation gaps.

Reproducibility Because this is a conceptual synthesis rather than a systematic review, the literature identification process is not fully reproducible in the systematic review sense. The targeted search strategy, search dates, databases, and inclusion criterion are described above; Table 1 documents the provenance of each identified mechanism. The analytical claim is that the six mechanisms are structurally distinct and collectively cover the major structural failure modes of threshold governance; the claim is not that they are exhaustive of all possible threshold-related concerns. Whether other analysts would derive the same set remains an empirical question that the proposed inter-rater reliability exercise (see Discussion) is designed to address. Regulatory alignment All regulatory alignment assessments are interpretive inferences by the author reflecting the governance intent of the cited provisions. They should not be read as statements of regulatory intent or legal requirement. Threshold classification Primary Safety Thresholds are defined as those where failure can directly cause patient harm.

Secondary Operational Thresholds govern operational performance where failure does not directly cause irreversible patient harm. When classification is uncertain, Primary Safety requirements apply by default. RESULTS: SIX STRUCTURAL FAILURE MECHANISMS Conceptual synthesis identified six structural failure mechanisms in threshold design. Table 1 documents the synthesis provenance.

Failure Mechanism Structural Description Key Sources Governance Gap 1. Proxy thresholding Metric bears only indirect causal relationship to the harm it purports to control Suresh & Guttag 2021 [16]; Wiens et al 2019 [17]; Char et al 2018 [15] Most frameworks accept metric achievement as sufficient; causal chain to harm not required 2. Context collapse Uniform thresholds applied across settings with materially different risk profiles Coiera 2019 [20]; Obermeyer et al 2019 [4]; Kelly et al 2019 [3]; FUTURE-AI [26] EU AI Act Annex IV requires deployment context but does not mandate threshold re-specification per context 3. Threshold coupling Multi-metric governance systems assess thresholds independently; compound failure modes remain undetected Leveson 2011 [7]; Rajkomar et al 2019 [1]; Parasuraman & Manzey 2010 [21] Least addressed in current standards; ISO 42001 references risk interaction but provides no methodology 4. Boundary gaming Hard cutoffs create incentives to optimise the metric rather than improve underlying safety Thomas & Uminsky 2022 [19]; Goodhart 1981

[18]; Raji et al 2020 [6] Algorithmic audit independence principles address gaming but are not mandated 5.

Epistemic asymmetry Quantifiable performance metrics crowd out harder-to-measure sociotechnical risks Selbst et al 2019 [13]; Ibrahim et al 2021 [28]; Vokinger et al 2021 [22] WHO 2021/2024 [14] addresses sociotechnical risks qualitatively; no quantitative governance counterpart 6. Audit–lifecycle disconnect Threshold justified at initial audit loses validity as population, data pipeline, or system version changes FDA PCCP 2023 [9]; Subbaswamy & Saria 2020; FDA SaMD 2021 [8]; FDA 2025 [27] FDA PCCP and EU AI Act Art. 72 address monitoring but do not require prespecified threshold re-evaluation triggers Table 1. Synthesis provenance: failure mechanisms, structural descriptions, literature sources, and governance gaps. Sources identified through targeted literature search (PubMed, Web of Science, Google Scholar; January 2018–December 2025) and citation tracking. This is a structured conceptual synthesis, not a systematic review. Governance gap column reflects the author's interpretive assessment. The Threshold Justification Stack: Specification and Tiered Requirements Six documentation elements are specified for TJS-compliant governance records. Table 2 presents the full specification and tiered requirements. TJS Layer Description Mechanism Addressed Tier Requirement Regulatory Counterpart 1. Harm-Control Statement Document the specific harm the threshold constrains and the causal pathway Epistemic asymmetry; proxy thresholding Required: Primary. Recommended: Secondary. EU AI Act Art. 9; FDA SaMD 2.

Context Key Specify clinical setting, patient population, and acuity level Context collapse Required: Both tiers EU AI Act Annex IV; FUTURE-AI [26] 3.

Calibration Evidence Empirical, simulation, or expert-consensus evidence supporting the numeric value Proxy thresholding; epistemic asymmetry Required: Primary. Optional: Secondary. FDA PCCP; CONSORT-AI 4.

Sensitivity Analysis

Evaluate governance outcomes as threshold varies; identify cliff-edge regions Threshold coupling; boundary gaming Required: Primary. Optional: Secondary. FMEA; ISO/IEC 42001 5. Gaming Resistance Assess whether the threshold can be cleared through metric manipulation Boundary gaming Required: Primary. Not required: Secondary. NIST AI RMF; audit independence 6.

Lifecycle Validity Rule Prespecify conditions triggering threshold re-evaluation Audit–lifecycle disconnect Required: Both tiers FDA PCCP; EU AI Act Art. 72 Table 2.

The Threshold Justification Stack: components, failure mechanisms addressed, tier requirements, and regulatory counterparts.

Comparison With Existing Governance Approaches Table 3 compares TJS threshold documentation requirements against those of existing governance instruments to illustrate that the TJS addresses gaps not covered by current practice.

Framework Harm-Control Context Key Calibration Sensitivity Gaming Lifecycle Tiered? DCB0129/0160 Partial (hazard log) Yes (intended use) No No No No No DTAC No Partial No No No Partial No NICE ESF No Partial Yes (Tier 3b) No No No Tiered by evidence level FUTURE-AI [26] Partial (fairness) Yes (universality) Yes (robustness) No No Partial (monitoring) No FDA PCCP [9] No Partial Partial No No Yes (change control) Risk-based TJS (this paper) Yes Yes Yes Yes Yes Yes Yes (Primary/Secondary) Table 3. Comparison of threshold documentation requirements across governance frameworks. All characterisations are the author's interpretive assessment. DCB0129: NHS Clinical Risk Management standard. DTAC: Digital Technology Assessment Criteria. NICE ESF: Evidence Standards Framework. FDA PCCP: Predetermined Change Control Plans.

Concrete gaming resistance pathways and assessment methodology are detailed in Supplementary Appendix F.

Integrating TJS Into Existing Hospital Audit Infrastructure TJS integration requires three procedural changes to existing AI governance processes: (1) adding a threshold classification field (Primary Safety / Secondary Operational / Uncertain) as the first element of any threshold submission; (2) populating the six TJS layers as required or optional fields according to tier; and (3) routing the completed TJS record to the governance committee alongside the existing AI assessment dossier.

The TJS field mapping to JSON/XML audit schemas and detailed integration specifications for EHR governance modules are provided in Supplementary Appendix C.

Interaction With Existing Hazard Logs and Clinical Risk Management Processes The TJS is designed to augment, not replace, existing NHS clinical risk management processes under DCB0129 and DCB0160 [30,31]. The clinical hazard log required under DCB0129 identifies potential hazards and assigns risk classifications; the TJS adds a structured documentation requirement for the governance thresholds that operationalise those risk classifications into deployment decisions. Specifically, L1 (Harm-Control Statement) maps to and extends the hazard identification function by requiring an explicit causal pathway from the threshold value to the harm it constrains. L4 (Sensitivity Analysis) extends the clinical risk assessment by documenting how governance outcomes change as the threshold varies—information not currently required in hazard log entries. L6 (Lifecycle Validity Rule) maps to the ongoing monitoring obligations under DCB0160 by prespecifying the conditions under which a threshold must be re-evaluated, linking static threshold documentation to the dynamic clinical safety management process. This positioning as “structured augmentation” rather than “new structure” is important: it reduces the perceived implementation burden by making clear that the TJS builds on governance artefacts already in use, rather than requiring institutions to adopt an entirely new framework.

Mapping of TJS layers to existing NHS governance artefacts is provided in Supplementary Appendix G.

Illustrative Committee Vignette An illustrative 45-minute committee review of a Primary Safety TJS record is described to illustrate that the review process is compatible with existing NHS governance cadence. A Clinical AI Oversight Committee receives a deployment proposal for an AI-assisted chest X-ray triage tool (AUROC ≥0.85, classified as Primary Safety). The committee reviews all six layers within its standing monthly meeting: L1 identifies the harm-control pathway (missed pathology leading to delayed diagnosis); L2 specifies the deployment context (emergency department, adult patients, urban teaching hospital); L3 presents calibration evidence from external validation; L4 indicates that performance degrades sharply below AUROC 0.80 but is stable between 0.85–0.95; L5 identifies no gaming vulnerability given independent external validation; and L6 specifies re-evaluation triggers (quarterly performance audit, population drift exceeding 15% from training distribution, or system version update).

Committee decision outcome: the committee approves deployment subject to two conditions: (1) prospective elderly-subgroup monitoring (patients aged 75+) at 90 days, with a prespecified performance floor of AUROC ≥0.80 in that subgroup; and (2) a scheduled formal review at six months coinciding with the routine clinical safety case update under DCB0160. The approval is recorded as “Approved with conditions” in the governance register, with the next review date of [date + 6 months] and the named accountable lead documented. If the 90-day subgroup audit reveals performance below the prespecified floor, the deployment reverts to “suspended pending investigation” status, triggering the L6 lifecycle re-evaluation process. This vignette is illustrative, not prescriptive; committee durations and compositions will vary. Its purpose is to demonstrate feasibility and to illustrate that TJS review produces a structured, auditable decision output compatible with existing governance cadence.

The six structural failure mechanisms converge on two fundamental sensitivity dimensions: the parameter calibration problem and the evidence calibration problem. Details are provided in Supplementary Appendix D.

Documentation Burden: Addressing the Proportionality Objection The tiered structure is the primary response to proportionality concerns: Secondary Operational Thresholds require only three of six layers, substantially reducing documentation effort.

For Primary Safety Thresholds, the documentation burden of all six layers is defended as proportionate to the irreversibility of their failure consequences. The investment is comparable to that already required for medical device regulatory submissions under the EU AI Act’s risk management provisions (Article 9) and the FDA’s PCCP framework. DISCUSSION

Contribution and Classification The TJS is classified as a normative governance proposal: it specifies what threshold documentation should contain, synthesised from existing safety engineering, regulatory, and algorithmic auditing principles, rather than reporting empirical findings about what threshold documentation does contain. Its practical contribution is to consolidate dispersed best-practice requirements into a single, tiered, audit-ready architecture. Its intellectual contribution is the argument that threshold governance failures are predominantly structural—arising from how thresholds are designed and documented—rather than from poor choices of individual numeric values. Relationship to FUTURE-AI and Recent Governance Literature The FUTURE-AI consensus guideline [26] defines lifecycle principles for trustworthy AI but does not specify how the quantitative thresholds within its 30 best practices should be justified or documented.

The TJS addresses this gap: it provides a structured documentation architecture for the numeric values that operationalise governance principles like fairness, robustness, and monitoring. Table 3 indicates that the TJS covers threshold documentation dimensions not addressed by any single existing framework.

The FDA's January 2025 draft guidance [27] reinforces several TJS layers—particularly L6 (Lifecycle Validity Rule), which maps to the total product lifecycle concept—but does not mandate the full six-layer documentation standard proposed here.

Tier Classification Reliability The TJS depends on consistent classification of thresholds as Primary Safety or Secondary Operational. Borderline cases will arise in practice. The "uncertain" classification option, which defaults to Primary Safety requirements, provides a conservative pathway for ambiguous cases. The reliability of tier classification across different assessors is an empirical question that this conceptual paper does not resolve. A recommended first step is an inter-rater reliability exercise (for which the unified core-equivalent dataset v4.1 provides standardised encoding templates with per-feature triangulation classification and explicit provenance annotation): two or more clinicians independently classify a sample of 10–15 thresholds currently in use within a single institution, using only the tier definitions and worked examples provided in this paper. This exercise is low-cost and would provide the first empirical evidence on the usability of the tier distinction.

Threshold Coupling: The Least Mature Layer Threshold coupling (Layer 4) remains the structural failure mechanism most inadequately addressed in current governance practice. The worked example in Supplementary Appendix A illustrates a practical coupling analysis for the AUROC–calibration floor interaction. Further methodological development of multi-threshold interaction governance is needed.

Regulatory Alignment: Scope and Limits The TJS layers map to requirements expressed in the EU AI Act, FDA PCCP guidance, FUTURE-AI, and WHO guidance (Table 2). However, all regulatory alignments are the author's interpretive inferences.

The EU AI Act does not explicitly mandate TJS-style threshold documentation. This paper argues that TJS-compliant documentation is consistent with the governance intent of these provisions, not that it is legally required.

Proposed Pilot Design With Prespecified Feasibility Endpoints A pilot within an existing hospital AI oversight committee is recommended as the first empirical step. The proposed pilot design specifies the following prespecified endpoints to enable evaluation of both usability and reliability. Primary endpoint: median completion time per TJS record, measured separately for Primary Safety (six-layer) and Secondary Operational (three-layer) threshold records. This metric directly addresses the proportionality concern: if six-layer completion time is prohibitive relative to existing clinical safety case preparation, the documentation burden is disproportionate. Secondary endpoints: (a) inter-rater agreement (Cohen's kappa) for tier classification (Primary Safety vs Secondary Operational vs Uncertain) across a sample of 10–15 thresholds, using only the tier definitions and worked examples provided in this paper; (b) committee acceptability, measured by Likert-scale rating and structured qualitative feedback (e.g. "which fields were hardest to complete?" and "which layers added most governance value?"); (c) comparison of TJS record completeness with current threshold documentation practices (proportion of TJS fields that are currently documented under existing governance processes). This pilot design does not fabricate results; it specifies what would be measured and how. The pilot is low-cost, requires no change to clinical care pathways, and is embeddable within existing AI oversight committee workflows. A kappa coefficient of ≥0.60 for tier classification is proposed as an illustrative feasibility benchmark, not a normative standard; it would provide preliminary evidence of acceptable reliability; values below this threshold would indicate that the tier definitions require refinement before broader adoption. A secondary analysis will explicitly evaluate decision stability under controlled joint perturbation of threshold parameters and evidence confidence bounds. This analysis is designed to characterise interaction effects between parameterisation and evidence quality under institutional conditions, enabling empirical mapping of the decision surface defined by these two dimensions.

Perturbation testing showed 90% ranking stability under ±20% variation. A threshold-sensitive region (0.40–0.45) produced binary outcome reversals. These observations are dataset-specific and non-generalised.

Consistent Signals From Historical Replay A companion historical replay evaluation of 12 documented AI governance failures using a non-compensatory (conjunctive) gate framework provides preliminary evidence relevant to threshold governance. In that study, the safety gate threshold was the most frequently binding constraint (83% of cases), while the evidence gate was least frequently binding (25%).

This pattern suggests that threshold calibration effort should be concentrated on safety and calibration dimensions, where threshold governance has the highest marginal impact.

The TJS framework provides the documentation architecture needed to justify such calibration priorities transparently.

Threshold sensitivity analysis using PhysioNet-derived governance vectors is described in Supplementary Appendix B. A cliff-edge region between 0.40 and 0.45 was identified for the safety gate.

The evidence enrichment and repository integration pipeline is described in Supplementary Appendix A.

This work represents a conceptual synthesis and normative governance proposal and does not constitute prospective real-world validation. The companion empirical programme (unified core-equivalent dataset v4.1, Evidence Registry v1) provides operationalisation artefacts—including standardised encoding templates with per-feature triangulation, provenance-tracked feature values, and confusion matrix outputs (TP=20, TN=12)—that can be reused as implementation evidence for the TJS methodology.

This paper contributes specifically by proposing the tiered Threshold Justification Stack as a proportionate documentation architecture for governance threshold governance.

The TJS is a normative proposal without empirical validation. The six failure mechanisms were identified through conceptual synthesis by a single author; whether other analysts would derive the same set is an empirical question that the proposed inter-rater reliability exercise is designed to address. The classification boundary between Primary Safety and Secondary Operational thresholds will require institutional deliberation in edge cases. Documentation burden of full six-layer compliance has not been tested in practice; the proposed pilot design with prespecified feasibility endpoints provides the pathway for this assessment. The threshold coupling layer (L4) is the least methodologically mature component. The gaming resistance layer (L5) specifies three concrete exploitation pathways, but additional pathways may be relevant in specific contexts. Interaction effects between TJS documentation and other governance processes have not been assessed, although the mapping to DCB0129/0160 hazard logs (above) indicates structural compatibility.

Companion studies provide preliminary, context-specific empirical signals consistent with the proposed governance framework, including 91.7% sensitivity (11/12) in historical replay of the canonical 12-case dataset under replay_mode, simulation evidence showing reduced unsafe approvals under non-compensatory gating, and 95.8% verdict stability under adversarial ±0.05 perturbation of calibration, bias, and traceability features (46/48 configurations stable; moderate profile fully invariant). Dual-dataset structural invariance testing confirmed that schema-level normalisation produces zero output divergence (480/480 fields identical). The canonical 12-case dataset is the primary validated dataset; expanded cases are secondary and exploratory. The perturbation dataset is synthetic and non-canonical.

CONCLUSIONS The tiered Threshold Justification Stack offers a proposed proportionate documentation architecture for linking governance thresholds to harm mechanisms, contextual conditions, and empirical evidence, with prespecified conditions for revision. The tiered structure—six layers for Primary Safety, three for Secondary Operational—addresses the proportionality concern while maintaining full requirements where threshold failure can directly cause patient harm.

The TJS is positioned as a structured augmentation of existing clinical risk management processes under DCB0129/0160, designed for integration into existing governance infrastructure rather than replacement of it. The worked examples in the supplementary appendices provide templates applicable within existing audit software, EHR governance modules, and clinical AI oversight committee workflows. The proposed pilot design with prespecified feasibility endpoints—including median completion time per threshold record and inter-rater agreement for tier classification—provides the necessary next step before claims of scalable applicability can be made. REFERENCES 1 Rajkomar A, Dean J, Kohane I. Machine learning in medicine. N Engl J Med 2019;380:1347-58. 2 Topol EJ. High-performance medicine: the convergence of human and artificial intelligence.

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No AI system generated the conceptual framework, threshold taxonomy, regulatory alignment mapping, worked examples, tier classification logic, hazard log interaction analysis, feasibility endpoint specification, or conclusions. All analytical content is the intellectual contribution of the author. What: Claude (Anthropic) generative AI tool. Why: to improve manuscript clarity, formatting consistency, and reference accuracy. How: sentence-level copy-editing and

formatting applied to author-drafted content; all outputs reviewed and verified by the author. Prompts and outputs are retained and can be supplied to editors on request. This disclosure is made in accordance with ICMJE recommendations and BMJ policy on the use of AI-assisted technologies.

Full reproducibility details are provided in Supplementary Appendix A.

Re-execution from identical inputs produces identical outputs. The corrected engine (corrected_public_engine_v1_1.py) is a deterministic transformation of the canonical governance engine, producing bit-identical results across runs. An earlier public engine implementation had drifted from canonical logic (incorrect 15-feature compensatory formula without bias inversion, threshold profile misalignment, gate evaluation anomalies); all results reported here use the corrected canonical logic (5-feature weighted compensatory model with bias inversion, locked threshold profiles). All results reported in the associated manuscripts were produced using the corrected engine (hash-locked implementation; see reproducibility artefact). Alternative implementations may produce different results.

Data Availability

No primary data were generated. The analytical framework is fully described in the manuscript; worked examples are provided in Supplementary Appendices A and B.

Contributorship Statement: WCB is the sole author and was responsible for all aspects of this work.

Ethics Statement

Ethical approval was not required. Patient and Public Involvement: This study did not involve patients or members of the public. Future empirical validation should include patient and clinician stakeholders. Supplementary Materials Supplementary Appendix A.

Full Worked Example: Six-Layer TJS Record (Primary Safety Threshold) — Emergency department sepsis prediction algorithm. Supplementary Appendix B.

Abbreviated Worked Example: Three-Layer TJS Record (Secondary Operational Threshold) — AI-driven clinical alert system. Supplementary Appendix C. TJS field mapping to JSON/XML audit schemas and EHR integration specifications.

Additional scope conditions and contextual material are detailed in Supplementary Appendix C.

Multimedia Appendix: PhysioNet Threshold Sensitivity

These patterns provide empirical motivation for the tiered TJS requirement that Primary Safety Thresholds receive all six documentation layers: the sensitivity of governance outcomes to small threshold changes at the safety gate indicates that this threshold's numeric value requires explicit calibration evidence (TJS Layer 3) and documented cliff-edge analysis (TJS Layer 4).